

WikiLLM – Intelligent Knowledge Management & Question Answering System

Project Overview

Build a production-ready AI-powered Knowledge Management System that transforms uploaded documents into a structured, interconnected wiki-style knowledge base instead of relying solely on Retrieval-Augmented Generation (RAG). The platform extracts entities, concepts, facts, relationships, summaries, and definitions, generating reusable wiki pages and a knowledge graph for advanced reasoning and question answering.

Primary Objectives

- Upload PDF, DOCX, TXT and Markdown files.
- Parse documents while preserving structure.
- Extract entities, concepts, definitions, facts, summaries and relationships using an LLM.
- Generate linked wiki pages automatically.
- Store knowledge in Markdown, JSON and PostgreSQL.
- Support semantic search, knowledge graph traversal and conversational question answering.

System Workflow

1. Document Upload
2. Document Parsing
3. LLM-based Knowledge Extraction
4. Wiki Page Generation
5. Knowledge Graph Creation
6. Knowledge Storage
7. Embedding Generation
8. Question Answering using structured knowledge plus semantic retrieval.

Core Features

Authentication, Dashboard, Document Management, Wiki Browser, Knowledge Graph Viewer, Full-text & Semantic Search, AI Chat Interface, Admin Panel, Multi-document Support, Persistent Knowledge Base.

Technology Stack

Backend: FastAPI (Python)
Frontend: React + TypeScript + Tailwind CSS
LLM Framework: LangGraph + LangChain
LLM: GPT-5 mini (or latest available)
Database: PostgreSQL
Embeddings: ChromaDB or pgvector
Knowledge Storage: Markdown + JSON
Document Parsing: PyMuPDF, python-docx

API Endpoints

POST /upload
GET /documents
GET /wiki
GET /wiki/{id}
POST /extract
POST /chat
POST /search
GET /graph

Non-Functional Requirements

Modular architecture, Docker-ready deployment, asynchronous processing, logging, error handling, OpenAPI documentation, unit tests, scalable design, clean architecture and SOLID principles.

Deliverables

Complete FastAPI backend, React frontend, PostgreSQL database, knowledge graph module, wiki generator, AI-powered question answering, REST API, Docker configuration, deployment guide, documentation, sample dataset, unit tests.